25901188

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER
Value Addition Course (VAC) - MG4VACBCA201 - IT AND ENVIRONMENTAL SUSTAINABILITY
(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Part A
Short Answer Questions
Answer any 7 questions
Each question carries 5 marks

1. Define biodiversity loss.

- Reduction in variety of plants, animals, and microorganisms
- Includes loss of species, genes, and ecosystems
- Caused by deforestation, pollution, climate change, overexploitation
- Leads to ecological imbalance
- Threatens food chain and human survival
- Results in extinction of species

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2. Describe the structure of the tundra ecosystem.

- Treeless cold biome found in polar regions
- Extremely low temperature and short growing season
- Soil remains frozen (permafrost)
- Vegetation includes mosses, lichens, grasses, small shrubs
- Animals include polar bears, reindeer, Arctic foxes
- Low biodiversity and low productivity
- Simple food chain due to harsh climate

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3. Differentiate between biotic and abiotic components with examples.

Biotic Components

- Living components of ecosystem
- Includes plants, animals, microorganisms
- Show growth and reproduction
- Examples: trees, humans, bacteria

Abiotic Components

- Non-living physical factors
- Includes air, water, sunlight, soil, temperature
- Influence survival of organisms
- Examples: rainfall, minerals, humidity

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4. Explain how sustainable practices help in reducing deforestation.

- Promotes afforestation and reforestation
- Encourages controlled use of forest resources
- Reduces dependence on wood through recycling
- Supports digitalization and paper reduction
- Promotes agroforestry and sustainable agriculture
- Creates awareness about forest conservation
- Helps maintain biodiversity and climate balance

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5. Describe the objectives of SDG 14 and explain why conserving marine life is essential for sustainable development.

Objectives of SDG 14 (Life Below Water)

- Conserve oceans and marine resources
- Reduce marine pollution
- Prevent overfishing
- Protect marine ecosystems and biodiversity
- Promote sustainable fishing practices
- Increase scientific knowledge of oceans

Importance of Conserving Marine Life

- Oceans regulate global climate
- Marine organisms support food security
- Source of employment and economy
- Maintains ecological balance
- Protects biodiversity
- Ensures sustainable use of ocean resources for future generations

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6. Analyze the causes of Delhi smog.

- Vehicular emissions causing air pollution
- Industrial smoke and factory emissions
- Burning of crop residue in nearby states
- Construction dust and road dust
- Firecrackers during festivals
- Winter temperature inversion trapping pollutants
- High population and urbanization
- Weak implementation of pollution control measures

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7. Explain how recycling conserves natural resources.

- Reduces need for raw materials
- Saves forests by recycling paper
- Conserves minerals and fossil fuels
- Reduces energy consumption in production
- Minimizes waste in landfills
- Decreases environmental pollution
- Supports sustainable resource management

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8. Why is sustainability necessary for development?

- Ensures resources for future generations
- Balances economic growth and environmental protection
- Reduces pollution and climate change
- Promotes social equality and quality of life
- Prevents depletion of natural resources
- Encourages responsible consumption
- Essential for long-term human survival and development

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9. Explain the relationship between society and the environment.

- Society depends on environment for resources
- Environment provides food, water, air, and shelter
- Human activities affect environmental quality
- Pollution and deforestation harm ecosystems
- Healthy environment supports human wellbeing

- Society has responsibility to conserve nature
- Both are interdependent and interconnected

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10. Explain the link between poverty and environmental degradation.

- Poor people depend heavily on natural resources
- Poverty leads to overuse of forests and land
- Lack of awareness increases pollution and waste
- Environmental degradation reduces agricultural productivity
- Natural disasters affect poor communities more
- Poor sanitation and waste management harm environment
- Creates a vicious cycle between poverty and degradation